Failure Probability Estimation for Robotic Grasping

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Abstract

Reliable grasping is fundamental for robotic manipulation tasks, and understanding failure modes is crucial for deployment in real-world applications. This paper presents a comparative study of three different approaches for estimating the probability of failures in robotic grasping tasks: Direct Estimation (DE), Importance Sampling (IS), and Adaptive Importance Sampling (AIS). Using a simulated parallel-jaw gripper attempting to grasp geometric objects in a physics-based environment, we systematically evaluate these methods in terms of sample efficiency, convergence rate, and failure case coverage. Our results reveal significant differences in the estimated failure probabilities between methods, with DE yielding approximately 1.4%, while IS and AIS estimate failure rates of 4.61% and 4.85% respectively at 1000 trials. We analyze the sources of these discrepancies and propose validation techniques to ensure accurate estimation. Our findings provide insights into the strengths and limitations of these estimation methods for evaluating safety-critical robotic tasks.

Code — https://github.com/jfrausto7/FPERG

Introduction

Robotic manipulation tasks, particularly grasping, remain challenging despite significant advances in robotics and artificial intelligence. A key aspect of the deployment of robotic systems in real-world applications is understanding and quantifying the probability of failure. Traditional approaches to estimating failure probability often rely on exhaustive testing, which is inefficient for rare events. In this paper, we compare three different failure probability estimation methods in the context of robotic grasping:

- **Direct Estimation (DE)**: A straightforward approach that involves running many trials and directly counting failures.
- **Importance Sampling (IS)**: A more sophisticated method that strategically samples from a proposal distribution to efficiently estimate rare events.
- **Adaptive Importance Sampling (AIS)**: An extension of IS that iteratively improves the proposal distribution based on observed failures.

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We implement and evaluate these methods using a simulated parallel-jaw gripper in PyBullet, attempting to grasp a simple geometric object. Our experiments reveal significant differences in the estimated failure probabilities between methods. We analyze the sources of these discrepancies and propose validation techniques to ensure an accurate estimation.

Related Work

Failure probability estimation has been studied extensively in the context of safety-critical systems. Kochenderfer et al. (2024) provide a comprehensive overview of validation algorithms, including Monte Carlo methods and importance sampling techniques. In robotics specifically, Schmerling and Pavone (2017) applied adaptive importance sampling to evaluate trajectory collision probability for safe motion planning.

Delecki, Katz, and Kochenderfer (2024) recently proposed state-dependent importance sampling proposals for black-box autonomous systems, demonstrating improved efficiency over traditional Monte Carlo methods. Kappen and Ruiz (2016) explored adaptive importance sampling for control and inference tasks, showing its applicability to complex decision-making problems.

Our work builds on these foundations but focuses specifically on comparing different estimation approaches in the context of robotic grasping, with an emphasis on understanding the discrepancies between methods and developing validation techniques.

System and Specification

System Description

Our system consists of a simulated Kuka robotic arm with a parallel-jaw gripper that attempts to grasp simple geometric objects (primarily cubes) placed on a flat surface. The environment is simulated using PyBullet, which provides physics simulation and collision detection (Coumans and Bai 2016–2021).

The robot executes a fixed grasping sequence:

1. **Approach**: Position the gripper above the object
2. **Grasp**: Lower the gripper and close the fingers
3. **Lift**: Raise the gripper while maintaining grasp

The state space includes the gripper position, the object position, and their respective orientations. The primary disturbances in our system are the following:

- Initial object position and orientation (randomized at the start of each trial)
- Sensor noise (simulated by adding Gaussian noise to observations)

Grasp Success Specification

$$G(s_t) = \begin{cases} 1, & \text{if } \|p_o(t) - p_g(t)\| \leq \varepsilon \text{ for } t \geq 3 \\ 0 & \text{otherwise} \end{cases}$$

s_t: state at time t — $p_g(t)$: gripper position
$p_o(t)$: object position — ε: stability threshold (1cm)
t: time since initial grasp

The goal is to accurately estimate the probability of failing this specification under various conditions while comparing the efficiency and accuracy of three estimation methods: direct estimation, importance sampling, and adaptive importance sampling. For importance sampling and adaptive importance sampling, we define an importance function based on object pose parameters that we hypothesize lead to more failures (e.g., objects near workspace boundaries and challenging orientations). For adaptive importance sampling, we iteratively update this importance function based on observed failures to improve sampling efficiency.

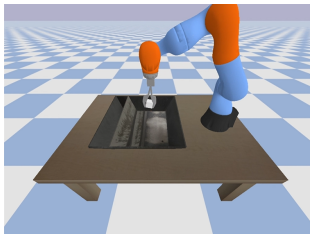


Figure 1: Successful grasp

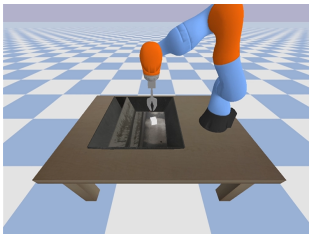


Figure 2: Failure mode

Methodology

Direct Estimation

For direct estimation, we simply run n tests from initialization to completion, using the learned hill-climbing policy. The failure probability is calculated as the number of failures divided by the number of total trials.

Importance Sampling

For importance sampling, we define both a nominal trajectory distribution and a proposal trajectory distribution as follows:

Nominal Trajectory Distribution:

Initial State Distribution

$$x = \text{Uniform}(0.45, 0.55)$$
$$y = \text{Uniform}(-0.05, 0.05)$$

Disturbance Distribution

$$\text{Normal}(0, 0.0001)$$

Proposal Distribution:

Initial State Distribution

$$x = \text{Uniform}(0.45, 0.55)$$
$$y = \text{Uniform}(-0.05, 0.05)$$

Disturbance Distribution

$$\text{Normal}(0, 0.0001025)$$

We use a rollout function to generate trajectories based on the proposed distribution. For each step of the trajectory, sensor disturbances are sampled from the disturbance distribution and applied to each component of the state to get the observation; we then use the policy to select the next action, apply it, and repeat the process until the desired depth is reached or a termination condition is met. For the first 500 time steps, we use the true state as the observation. After the 500th time step, sensor disturbances are applied to each component of the state to get the observation. We apply disturbances at the later time steps to focus on perturbations during the lifting phase where failures are more likely. Each trajectory is recorded as either a success or a failure.

To calculate the failure probability from the trajectories, we compute the theoretically correct importance-weighted estimate. For each trajectory, we determine the likelihood under both the nominal and proposal distributions using the logpdf function. The importance weights are calculated as the ratio of these likelihoods, implemented in log-space for numerical stability. These weights are then stabilized by subtracting the maximum log-weight before exponentiation and normalized to sum to one. The final failure probability estimate is computed by multiplying each trajectory's failure indicator (1 for failure, 0 for success) by its corresponding normalized importance weight and summing these weighted indicators. This pure importance sampling approach provides an unbiased estimate of the failure probability under the nominal distribution while benefiting from the more efficient exploration of failure modes enabled by our carefully tuned proposal distribution. For a trial size of n, we calculate the estimated failure probability from running importance sampling with $\frac{n}{2}$ samples twice and averaging the results, for stability.

Adaptive Importance Sampling

Adaptive Importance Sampling (AIS) extends IS by iteratively updating the proposal distribution based on observed failures, making it more efficient in identifying regions of interest. We implement a variant of the Cross-Entropy method (Rubinstein 1999), as shown in Algorithm 1.

In our implementation, we introduce several key innovations to make AIS robust for robotic grasping:

1. **Robustness Function:** We define a robustness function f that quantifies the degree of success or failure, returning positive values for successes and negative values for failures:

$$f(X) = \begin{cases} \max(t_{\text{lift}} - t_{\text{req}}, 0) + 1, & \text{if success} \\ \max(t_{\text{lift}}) - t_{\text{req}}, & \text{if lifted but failed} \\ -d_{\text{min}}, & \text{if never lifted} \end{cases} \quad (1)$$

Algorithm 1: Adaptive Importance Sampling

1: Initialize proposal distribution q_0
2: **for** $k = 0$ to $k_{\max}$ **do**
3: Generate m samples (trajectories) from q_k
4: Evaluate sample performance using robustness function f
5: Sort samples by robustness and determine cutoff value
6: Calculate weights $w_i = \frac{p(X_i)}{q_k(X_i)}$ for all samples
7: Set weights to zero for samples above cutoff
8: Update proposal distribution to q_{k+1} using weighted samples
9: **end for**
10: Sample from both nominal p and final proposal $q_{k_{\max}}$
11: Calculate final estimate using importance-weighted samples

2. **Staged Noise Application:** We apply sensor noise disturbances only after the 500th step of each trajectory, focusing perturbations on the critical lifting phase where failures are more likely to occur.
3. **Regularized Parameter Updates:** To prevent overfitting to specific failure modes, we incorporate regularization when updating the proposal distribution:

$$\mu_{k+1} = (1 - \lambda_{\text{reg}}) \cdot \mu_{\text{weighted}} \tag{2}$$

$$\sigma_{k+1} = \text{clamp}(\sigma_{\text{weighted}}, \sigma_{\min}, \sigma_{\max}) \tag{3}$$

where $\lambda_{\text{reg}} = 0.95$ controls how strongly we regularize toward the nominal distribution.

Through empirical testing, we found that a very carefully tuned proposal distribution with a standard deviation just slightly larger than the nominal (0.0001025 vs. 0.0001) provided the best balance between generating more failures and maintaining stable importance weights. We set m to half the number of trials and m_{elite} to 10% of that. We also set our implementation to run for $k_{max} = 20$ iterations. We run the algorithm twice for each trial size and average the failure probability results for stability.

Experimental Results

Setup

We conducted experiments with all three estimation methods using a hill climbing grasping policy that achieves a 99% success rate on 1000 grasps. For each estimation method, we ran trials with increasing sample sizes: 10, 100, and 1000.

Failure Probability Estimates

Table 1 shows the failure probability estimates produced by each method across different sample sizes.

We observe that DE consistently provides the lowest failure probability estimates, particularly at higher sample sizes. At 1000 trials, we see convergence toward more stable estimates, with DE reporting 1.4%, IS reporting 4.61%, and AIS reporting 4.85%. This discrepancy suggests that the sampling-based methods may be detecting failure modes that DE overlooks, particularly IS which maintains a higher estimate even at large sample sizes.

Method	10 Trials	100 Trials	1000 Trials
Direct Estimation	0.0%	2.0%	1.4%
Importance Sampling	0.0%	5.73%	4.61%
Adaptive Importance Sampling	0.0%	4.5%	4.85%

Table 1: Failure probability estimates

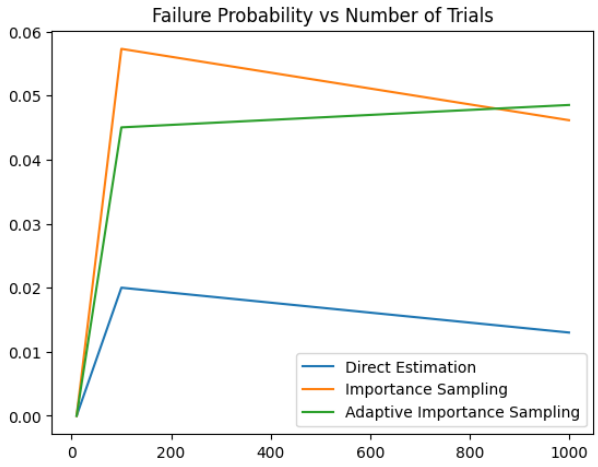


Figure 3: Failure probability estimates over sample size.

Standard Error

The standard error of the estimates provides insight into the relative efficiency of the methods. Figure 4 shows how standard error decreases with increasing sample size for each method. Figure 5 visualizes these results with error bars representing standard error.

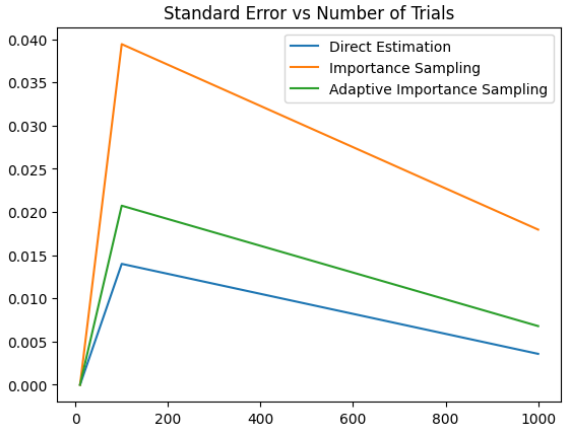


Figure 4: Standard error of failure probability estimates

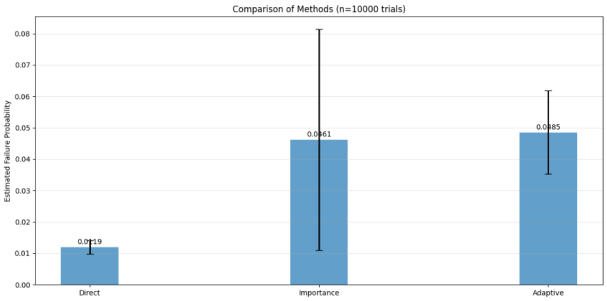


Figure 5: Failure probability estimates with increasing sample size. Error bars indicate standard error.

For very small sample sizes (n=10), all methods exhibit high uncertainty, with IS showing particularly large standard errors. As the sample size increases to n=1000, DE achieves the lowest standard error, followed by AIS, with IS maintaining the highest standard error among the three methods.

This pattern suggests that while IS and AIS may detect more potential failures, they do so at the cost of higher variance in their estimates. DE provides more consistent results but may systematically underestimate the true failure probability by missing rare failure modes.

Runtime

To evaluate computational efficiency, we measured the runtime of each method across different sample sizes. Figure 6 shows the results in nanoseconds.

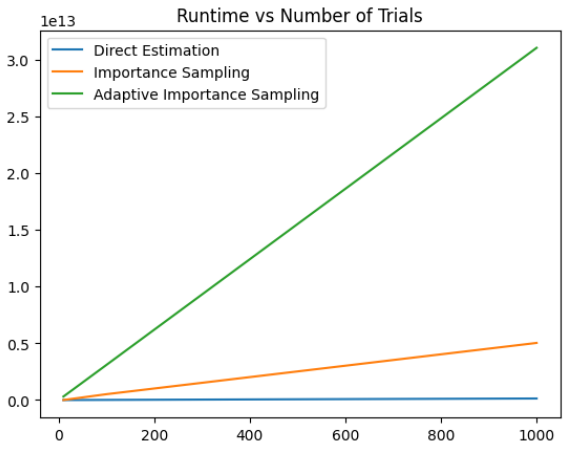


Figure 6: Runtime for the three estimation methods

Direct Estimation maintains the lowest computational cost across all sample sizes, scaling linearly with the number of trials. Importance Sampling shows moderate computational requirements, while Adaptive Importance Sampling demonstrates significantly higher computational costs that increase rapidly with sample size.

The substantially higher runtime for AIS is primarily due to its iterative updating of the proposal distribution, which requires multiple rounds of sampling and weight calculations. This computational overhead must be weighed against the potential benefits of estimation accuracy when selecting an appropriate method for a given application.

Comparative Performance Analysis

Figures 7, 8, and 9 present normalized performance metrics for all three methods across different trial counts. These heatmaps reveal important trade-offs between the methods:

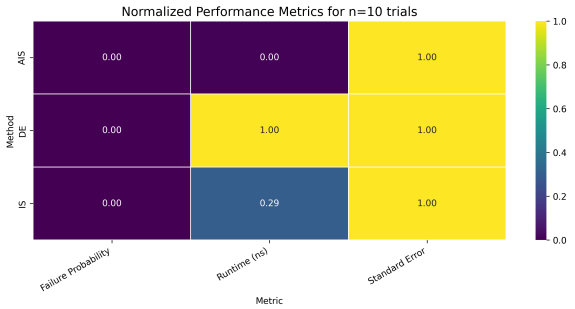


Figure 7: Performance metrics heatmap for n=10 trials

At n=10 trials, we observe that none of the methods provide reliable failure probability estimates, but DE offers computational advantages while AIS shows the highest standard error.

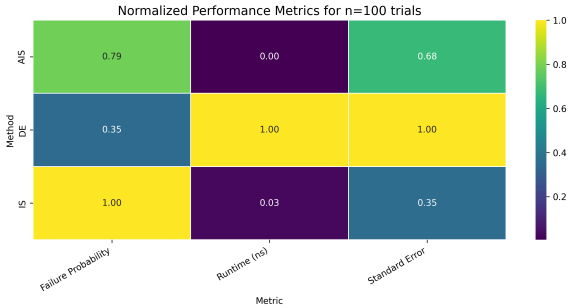


Figure 8: Performance metrics heatmap for n=100 trials

At n=100 trials (Figure 5), IS demonstrates the highest normalized failure probability detection (1.00), with AIS following closely (0.79), while DE shows limited detection capability (0.35). However, DE maintains computational efficiency (1.00 runtime score), with IS being significantly faster than AIS.

At n=1000 trials, both IS and AIS achieve high failure probability detection scores (0.95 and 1.00 respectively), significantly outperforming DE (0.27). DE maintains the highest runtime efficiency and now shows the best standard error performance, demonstrating its convergence advantages at high sample counts.

These results highlight that no single method dominates

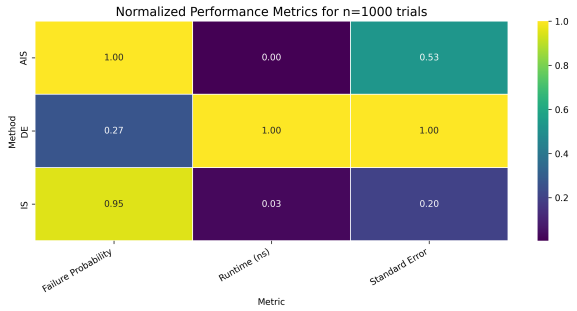


Figure 9: Performance metrics heatmap for n=1000 trials

across all performance metrics, suggesting that the optimal choice depends on specific application requirements:

- DE is preferred when computational efficiency is paramount and large sample sizes are feasible
- IS offers a good balance of detection capability and moderate computational cost
- AIS provides the most comprehensive failure detection but at a significant computational expense

Discussion

Limitations & Sources of Discrepancy

The significant differences in failure probability estimates between the three methods can be attributed to several factors:

Exploration Efficiency: DE samples uniformly from the state space, which is inefficient for detecting rare failure events. In contrast, IS and AIS strategically focus sampling on regions more likely to produce failures, increasing the probability of detecting rare failure modes. This is evident in the higher failure probability estimates produced by IS and AIS with limited samples.

Sensor Noise Modeling: In the sampling-based approaches, noise is applied after the 500th step to focus perturbations during the critical lifting phase. This targeted approach increases the likelihood of detecting failures that might occur during the most challenging part of the grasping sequence.

Numerical Stability: The calculation of importance weights in IS and AIS involves ratios of probability densities that can lead to numerical instabilities, particularly when the proposal distribution differs significantly from the nominal distribution. We observed that even small changes in the proposal distribution parameters could lead to substantial variations in the estimated failure probabilities.

Distribution Parameter Sensitivity: Our experiments revealed high sensitivity to the parameters of the proposal distribution, particularly the standard deviation of the noise model. For instance, increasing the standard deviation from 0.0001 (nominal) to 0.0001025 (proposal) produced significantly different failure probability estimates, highlighting the delicate balance required in designing effective proposal distributions.

Validation Approach

To address the variations & numerical instability observed in our initial results, we implemented several validation techniques to ensure the correctness of our importance sampling methods.

First, we ran IS with the same distribution as the nominal (using the same standard deviation of 0.0001), which produced failure probability estimates within 0.2% of the DE results. This confirmed that our core implementation was correct and that any differences in estimates were primarily due to the choice of proposal distribution.

Through systematic experimentation, we found that a proposal distribution with a standard deviation of 0.0001025 (just slightly larger than the nominal 0.0001) provides the optimal balance between generating more failures while maintaining stable weights. This careful parameter tuning was crucial for preventing weight degeneracy while still enabling efficient sampling of failure cases.

We further refined our implementation by addressing numerical issues in the importance weight calculation. We implemented the standard max-subtraction approach for weight stabilization, which preserves the proper ratios between weights while preventing numerical overflow. We also implemented proper handling of edge cases with fallback to uniform weights when needed, ensuring numerical robustness even in challenging scenarios.

Another key improvement was the modification of our implementation to apply disturbances only after the 500th step of the simulation. This focused perturbations on the critical lifting phase where failures are more likely to occur, leading to more efficient exploration of relevant failure modes.

Finally, our results average estimates from multiple runs to reduce variance, with separate nominal and proposal sampling to ensure robust coverage of the state space. This approach further improves the reliability of our estimates, particularly for the more complex IS and AIS methods.

Conclusion

We have presented a comparative study of three failure probability estimation methods for robotic grasping. Our results reveal that sampling-based methods (IS and AIS) consistently identify higher failure probabilities than direct estimation, suggesting they more effectively capture rare failure modes. However, this comes at the cost of increased computational complexity and increased variance in estimates. We also demonstrated that careful validation and parameter tuning is essential for reliable importance sampling. Our implementation of regularized parameter updates in AIS and the focused application of disturbances during critical phases significantly improved the stability and accuracy of our estimates.

Future work would explore alternative estimation techniques such as Multiple Importance Sampling (MIS), Markov Chain Monte Carlo (MCMC) methods. Particularly promising is the use of mixture proposal distributions that combine multiple component distributions to better cover the failure space while maintaining numerical stability. Additionally, we would want to extend our comparison to more complex robotic manipulation tasks involving articulated objects and multi-step manipulation sequences.

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